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1.0 INTRODUCTION

1.1 Purpose

This document outlines the comprehensive Test Plan for the Insurance Core Platform, covering Policy Administration, Claims Management, Billing & Premium Calculation, Underwriting, and Customer/Agent Portals. It defines testing objectives, strategy, scope, environments, schedules, and acceptance criteria aligned with insurance industry standards.

1.2 Background

The platform supports multiple Lines of Business (LOBs): Personal Auto, Homeowners, Commercial Property, Term Life, and Group Health. The system integrates with external rating bureaus, payment gateways, reinsurers, and regulatory reporting engines. Ensuring accuracy in premium/rating calculations, FNOL (First Notice of Loss) workflows, claims adjudication, and data privacy (HIPAA, GDPR, ACORD) is critical.

1.3 Objectives

- Validate end-to-end policy lifecycle (Quote → Bind → Issue → Endorse → Renew → Cancel)
- Verify claims processing accuracy (FNOL, Reserve Calculation, Payment, Recovery/Subrogation)
- Ensure premium/billing calculations align with actuarial rating engines
- Confirm regulatory compliance, data masking, and security controls
- Achieve 95%+ functional test coverage and <0.5% critical defect leakage to UAT

- # 2.0 TEST STRATEGY (IMPORTANT)
- ## 2.1 Testing Levels & Scope Coverage

Level	Focus Area	Tools/Techniques
Unit Testing	Rating engine formulas, validation rules, API contracts	JUnit, pytest, SonarQube, Code Coverage
Integration Testing	ACORD XML/JSON messaging, Payment gateways, Reinsurance feeds, E-signature	Postman, SOAP UI, WireMock, Contract Tests
System Testing (SIT)	End-to-end policy lifecycle, Claims adjudication, Billing cycles, UI/UX	Selenium, Playwright, Cypress, BDD (Cucumber)
UAT	Business rule validation, Agent/Customer workflows, Regulatory reporting	Manual + Exploratory, Real-world test data
Performance	Concurrent quote generation, Batch rating, Claims bulk upload	JMeter, Gatling, LoadRunner
Security & Compliance	PII/PHI masking, Role-based access, Audit trails, Penetration testing	OWASP ZAP, Burp Suite, DataMask Pro

2.2 Automation Strategy

- **Framework:** Hybrid (Keyword + Data-Driven + BDD) using Playwright + Python/Java
- **CI/CD Integration:** Jenkins/GitLab CI with nightly regression suite execution
- **Test Data Management:** Synthetic data generation with insurance-specific scenarios (e.g., multi-vehicle policies, partial cancellations, subrogation claims, grace period billing)
- **Automation Coverage Target:** 70% of stable regression paths, 100% API contract tests
- **Maintenance:** Page Object Model (POM) for UI, API abstraction layers, self-healing locators for dynamic insurance portals

2.3 Insurance-Specific Testing Focus Areas

Module	Key Validation Points
Policy Administration	Quote accuracy, binding rules, endorsement processing, policy state transitions, cancellations/refunds
Rating Engine	Tiered pricing, discounts/credits, territorial factors, driver/vehicle risk scoring, tax & fee calculations
Claims Management	FNOL intake, coverage verification, adjuster assignment, reserve calculations, payment scheduling, recovery tracking
Billing & Premiums	Installment plans, grace periods, late fees, proration, agency commission splits, EFT/CC processing
Underwriting	Risk rules engine, referral routing, manual underwriter queues, audit trail compliance

Module	Key Validation Points
Customer/Agent Portal	Self-service policy changes, document generation, claim status tracking, secure messaging

2.4 Test Data Strategy

- **Production-Like Data:** Masked/anonymized PII/PHI using tokenization
- **Scenario Coverage:** Valid/invalid risk profiles, boundary rating ages, multi-LOB bundles, historical claim patterns
- **Data Refresh:** Weekly refresh from sanitized production snapshot + synthetic edge-case injection
- **Compliance:** Adherence to NAIC, HIPAA, GDPR, and state-specific insurance regulations

2.5 Defect Triage & Prioritization

- Severity mapped to insurance impact (e.g., Incorrect premium calculation = Critical; UI typo on FAQ = Low)
- Daily triage meetings with QA, Dev, Actuarial, Compliance, and Business stakeholders
- SLA: Critical defects resolved within 24h, High within 72h

3.0 TEST SCOPE

3.1 In-Scope

- Policy Quoting, Binding, Issuance, Endorsements, Renewals, Cancellations
- Claims FNOL, Investigation, Adjudication, Reserving, Payments, Subrogation
- Billing Engine, Premium Calculations, Installments, Refunds, Commissions
- Customer Portal, Agent Portal, Underwriter Workbench
- Integration with Rating Bureaus, Payment Processors, E-Signature, Reinsurers
- Regulatory Reporting (State DOI filings, ACORD standards, Audit Logs)

3.2 Out-of-Scope

- Legacy Mainframe decommissioning validation (handled in separate migration plan)
- Third-party Actuarial Model development & mathematical certification
- Hardware/Infrastructure load testing (managed by DevOps/Cloud team)
- Marketing/Campaign system testing

4.0 TEST APPROACH & METHODOLOGY

4.1 Methodology

- Agile/Scrum aligned testing with shift-left practices

- BDD (Behavior-Driven Development) for business rule clarity: Given/When/Then format mapped to policy/claims workflows
- Traceability Matrix linking Requirements → Test Cases → Defects → Sign-off

4.2 Test Design Techniques

- Equivalence Partitioning & Boundary Value Analysis (for rating tiers, age limits, coverage amounts)
- State Transition Testing (Policy: Active → Lapsed → Reinstated → Cancelled)
- Decision Table Testing (Underwriting rules, Discount eligibility, Claims coverage validation)
- Pairwise Testing (Multi-coverage combinations, Deductible × Limit matrices)

4.3 Regression Strategy

- Automated smoke suite executed on every build
- Full regression executed pre-UAT and pre-Production deployment
- Impact analysis-driven selective regression for hotfixes

5.0 TEST ENVIRONMENT & TOOLS

Environment	Purpose	Data State	Access Control
DEV	Unit & component testing	Mock/Synthetic	Dev/QA
SIT	Integration & system testing	Masked Production	QA/BA
UAT	Business validation	Sanitized Prod-like	Business/Users
PERF	Load/Performance testing	Synthetic bulk	QA/Perf Engineers
STAGE/PROD	Final validation & smoke testing	Production replica	Release/Compliance

5.2 Toolchain

- **Test Management:** JIRA Xray / TestRail
- **Automation:** Playwright, Selenium, Postman, RestAssured, Pytest
- **Performance:** JMeter, k6
- **Security:** OWASP ZAP, Snyk, Veracode
- **Data Masking:** IBM InfoSphere / TokenEx
- **CI/CD:** Jenkins, GitLab CI, Docker, Kubernetes
- **Monitoring:** Splunk, Datadog, ELK Stack

6.0 TEST DELIVERABLES

- Approved Test Plan & Test Strategy
- Requirement Traceability Matrix (RTM)
- Test Cases & Test Scripts (Manual + Automated)
- Test Data Sets (Masked & Synthetic)
- Execution Reports (Daily/Weekly/Sprint)
- Defect Reports & Trend Analysis
- Performance & Security Test Reports
- UAT Sign-Off & Release Readiness Report
- Lessons Learned & QA Metrics Dashboard

7.0 SCHEDULE & MILESTONES

Phase	Duration	Start Date	End Date	Deliverable
Test Planning & Strategy	10 Days	Apr 13, 2026	Apr 24, 2026	Approved Test Plan & RTM
Test Case Design	12 Days	Apr 27, 2026	May 12, 2026	Test Cases & Automation Scripts
SIT Execution	15 Days	May 13, 2026	Jun 02, 2026	SIT Reports & Defect Log
UAT Preparation	5 Days	Jun 03, 2026	Jun 09, 2026	UAT Test Pack & Environment
UAT Execution	10 Days	Jun 10, 2026	Jun 23, 2026	UAT Sign-Off
Regression & Go/No-Go	5 Days	Jun 24, 2026	Jun 30, 2026	Release Readiness Report

8.0 RISKS & MITIGATION

Risk	Impact	Probability	Mitigation Strategy
Rating engine formula discrepancies	High	Medium	Parallel run with actuarial validation, automated precision checks (±0.01 tolerance)
PII/PHI data exposure in non-prod	Critical	Low	Strict data masking, tokenization, quarterly compliance audits
Third-party API downtime (Payment/Reinsurer)	High	Medium	Mock services, circuit breaker patterns, fallback workflows, SLA monitoring
Regulatory rule changes mid-cycle	High	Medium	Configurable rules engine, change control board, rapid test patch deployment

Risk	Impact	Probability	Mitigation Strategy
Test environment instability	Medium	High	Infrastructure-as-Code (IaC), automated health checks, dedicated QA env ownership

9.0 ROLES & RESPONSIBILITIES

Role	Responsibilities
QA Lead	Test strategy, planning, reporting, stakeholder alignment, risk management
Test Engineers	Test case design, execution, automation scripting, defect logging
Business Analysts	Requirement clarification, UAT coordination, business rule validation
Actuarial/Compliance SME	Rating accuracy validation, regulatory compliance sign-off, audit trail review
DevOps/Environment Owner	Environment provisioning, CI/CD pipeline, data refresh, tool access
UAT Testers (Business)	End-to-end workflow validation, real-world scenario execution, final sign-off

10.0 ENTRY & EXIT CRITERIA

10.1 Entry Criteria (SIT/UAT)

- Requirements baselined & traceable
- Test environment provisioned & stable
- Test data masked & loaded
- Critical/High defects from previous phase resolved
- Build passes smoke test (100% critical paths)

10.2 Exit Criteria (Release)

- 100% test cases executed
- ≥95% Pass rate
- Zero Critical/High open defects
- Performance meets SLA (≤2s response for 95th percentile, 500 concurrent users)
- Security scan passed with no Critical/High vulnerabilities
- UAT sign-off received from Business & Compliance
- Regression suite green in CI/CD pipeline

11.0 DEFECT MANAGEMENT

11.1 Severity & Priority Matrix

Severity	Description	Resolution SLA	Priority Mapping
Critical	System crash, incorrect premium/claims payout, data breach	24h	P1
High	Core workflow blocked, major calculation error, compliance gap	72h	P2
Medium	Workaround available, UI/UX defect, minor integration delay	5 Days	P3
Low	Cosmetic, documentation, non-blocking	Next Sprint	P4

1.2 Workflow

New → Assigned → In Progress → Fixed → Verified → Closed / Reopened
All defects require steps to reproduce, test data, logs, and screenshots/video.

12.0 APPROVALS & SIGN-OFF

Role	Name	Signature	Date
QA Lead			
Project Manager			
Product Owner			
Compliance/Actuarial			
Release Manager			

APPENDIX A: INSURANCE-SPECIFIC COMPLIANCE MATRIX

Regulation/Standard	Coverage Requirement	Test Validation Method
HIPAA/GDPR	PII/PHI encryption, consent management, DSR	Data masking verification, audit log review
NAIC Model Laws	Policy language, cancellation notices, grace periods	Document generation, workflow timing tests
ACORD Standards	XML/JSON messaging, form layouts	Schema validation, integration contract tests

Regulation/Standard	Coverage Requirement	Test Validation Method
State DOI Rules	Rate filing compliance, surplus lines reporting	Business rule engine validation, reporting audits

APPENDIX B: TEST CASE COVERAGE SUMMARY

Module	Test Cases	Automated	Manual	Pass Rate Target
Policy Administration	420	290	130	96%
Claims Management	380	250	130	95%
Billing & Premiums	310	220	90	97%
Underwriting	190	110	80	94%
Portals (Customer/Agent)	280	190	90	95%
TOTAL	**1,580**	**1,060**	**520**	**≥95%**